# Data Complexity via Data Geometry Using Diffusion Models

Score-induced metric · heat kernel · $\log\det(I+K)$

Kirill Zolotarev, MIPT

May 2026

# Problem Statement

## Task

Given a sample $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^D$ near a manifold $\mathcal{M}$, define a **scalar complexity** $C(X)$ that reflects intrinsic geometry, not only sample size $n$.

## Problem

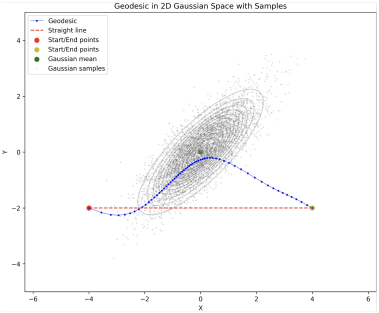
Classical approaches (PCA, local intrinsic-dimension estimators) are either linear or purely local, and do not produce a **global** geometric functional. Diffusion models learn $\mathbf{s}_\theta(x) \approx \nabla \log p(x)$ — we want to turn this learned field into one such functional.

# Related Work

- **Score & manifold ID** — Stanczuk et al., 2023
  Score aligns with normal bundle at small noise; intrinsic dimension from diffusion model.
- **Geometric complexity from diffusion** — Kamkari et al., 2024
  Local complexity probes via learned generator geometry.
- **Score-induced metric** — Perone, 2024
  $g = (I + ss^\top)^{-1}$ links Stein score to ambient Riemannian structure.
- **Diffusion maps** — Coifman & Lafon, 2006
  Graph Laplacian + diffusion distances from local kernels; geometry from similarities.
- **Local kernels / Berry–Sauer** — Berry & Sauer, 2016
  Double normalization recovers $\Delta_g$ on manifolds from ambient samples (Thm. 4.7).

*Our angle:* couple score metric with Berry–Sauer heat kernel and $\log\det(I+K)$ for a dataset-level functional with algebraic guarantees.

# Central Idea




*Induced geodesics, Perone, 2024*

1. **Score.** $\mathbf{s}(x) = \nabla_x \log p_\tau(x) \approx \mathbf{s}_\theta(x)$
2. **Metric.** $g(x) = \left(I + \mathbf{s}(x)\mathbf{s}(x)^\top\right)^{-1}$ contracts along $\mathbf{s}$.
3. **Laplace–Beltrami operator.** $\Delta_g$ on $(\mathcal{M}, g)$.
4. **Heat kernel + approximation.**

$$h_t(x,y) = \sum_{\ell \geq 0} e^{-t\lambda_\ell} \phi_\ell(x)\phi_\ell(y), \qquad H_t = \exp(-t\, L_\epsilon^{\mathrm{sym}})$$

$L_\epsilon^{\mathrm{sym}} \to \Delta_g$ via local-kernel / Berry–Sauer normalization.

**Asymptotic intuition.**

$$h_t(x,y) \asymp (4\pi t)^{-d/2} \exp\!\left(-d_g(x,y)^2/4t\right), \quad t \to 0^+$$

strong coupling along short $g$-geodesics through high-density regions.

5. **Complexity.** $C(X) = \log\det(I + K), \quad K \succeq 0.$

# Method — Discrete Construction

1. **Local anisotropic kernel** on the sample, with shape matrix $A(x) = g(x)^{-1} = I + \mathbf{s}\mathbf{s}^\top$:

$$K_\epsilon(x_i, x_j) = \exp\Big(-\tfrac{1}{2\epsilon}(x_i - x_j)^\top g(x_i)(x_i - x_j)\Big), \quad \bar{K}_\epsilon = \tfrac{1}{2}(K_\epsilon + K_\epsilon^\top).$$

2. **Berry–Sauer double normalization**:

$$q_\epsilon(x_i) = \sum_j \bar{K}_\epsilon(x_i, x_j), \quad \tilde{K}_\epsilon = \frac{\bar{K}_\epsilon(x_i, x_j)}{q_\epsilon(x_i) q_\epsilon(x_j)}, \quad d_\epsilon(x_i) = \sum_j \tilde{K}_\epsilon(x_i, x_j).$$

$$L_\epsilon^{\mathrm{sym}} = \tfrac{2}{\epsilon}\big(D_\epsilon^{-1/2} \tilde{K}_\epsilon D_\epsilon^{-1/2} - I\big).$$

Result: score-induced metric fits the framework

3. **Spectral construction of the heat kernel.** Diagonalize the symmetric $L_\epsilon^{\mathrm{sym}} = \sum_k \mu_k \phi_k \phi_k^\top$ ($\mu_k \le 0$), then

$$H_t = \exp(t\, L_\epsilon^{\mathrm{sym}}) = \sum_{k=1}^{n} e^{t\mu_k} \phi_k \phi_k^\top \succeq 0.$$

4. **Complexity functional.** $C(X; t, \epsilon) = \log\det(I + H_t) = \sum_{k=1}^{n} \log\big(1 + e^{t\mu_k}\big).$

# Properties (theorems in the paper)

Under symmetry and $K \succeq 0$ (normalized discrete heat kernel), the complexity $C(X) = \log\det(I+K)$ satisfies:

- **Theorem (basic):** permutation invariance; invariance under Euclidean rigid motions $x \mapsto Rx + t$ (score transforms covariantly).
- **Theorem (bounded growth):** if $K_{ii} = 1$, then $0 \le C(X) \le n \log 2$ and each marginal increment lies in $[0, \log 2]$.
- **Theorem (monotonicity):** $A \subseteq B \Rightarrow C(A) \le C(B)$.
- **Theorem (subadditivity):** for disjoint $A, B$, $C(A \sqcup B) \le C(A) + C(B)$.
- **Theorem (submodularity):** $C(X \cup \{x\}) - C(X) \ge C(X' \cup \{x\}) - C(X')$ for $X \subseteq X'$, $x \notin X'$
- **Theorem (Lipschitz stability):** $|C(K) - C(K')| \lesssim \sqrt{n}\,\|K - K'\|_F$ for PSD kernels $K, K'$.

PSD of $K$ is proved separately

# Computational Experiments — Overview

**Four verification tracks:**

1. **Exp. 1 — $S^2$, analytic.**
   Closed-form $h_t$ on the sphere. Verify bounds $\log(1+n) \le C \le n \log 2$, monotonicity, subadditivity.
2. **Exp. 2 — $S^2$, discrete vs. analytic.**
   Berry–Sauer $H_t$ on a uniform sample converges to analytic $h_t$; functional $C^{\text{approx}} \to C^{\text{oracle}}$.
3. **Exp. 3 — $\mathbb{R}^2$ Gaussian mixture.**
   Analytic score; $C$ detects number of clusters and their separation; $K$ shows block structure.
4. **Exp. 4 — noisy torus in $\mathbb{R}^3$.**
   Trained score; score-induced kernel recovers intrinsic geometry where the Euclidean baseline fails.

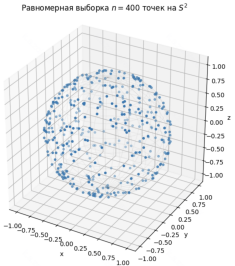
**Common setup:**

- bandwidth $\epsilon$: chosen via the $\log T(\epsilon)$ plateau (Coifman–Lafon diagnostic);
- time $t$: swept on a log-grid; reported at $t_{\text{med}}$ and at the spectral gap;
- baseline: same pipeline with $g{=}I$ (Euclidean kernel).

# Experiment 1 — $S^2$: bounds & redundancy

**Goal.** Validate $C(X) = \log\det(I+K)$ between $\log(1+n)$ (duplicates) and $n\log 2$ (spread), using analytic spherical $h_t$ with diagonal normalization.

Few points on $S^2$




Many points on $S^2$

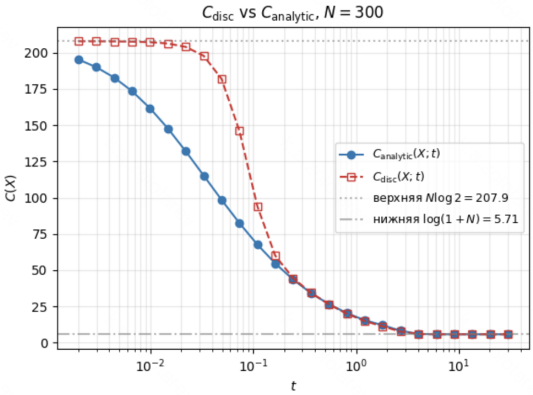
plot0.png

$C(n)$ vs. bounds $\log(1+n)$, $n\log 2$

Duplicate configurations saturate the lower bound $\log(1+n)$; spread configurations approach the upper bound $n\log 2$.

# Experiment 2 — $S^2$: analytic vs. discrete heat

**Goal.** On the same sample, compare $C(X;t) = \log\det(I+K_t)$ built from the continuum heat kernel and from the Berry–Sauer discrete semigroup.




$C_{\text{analytic}}(t)$ vs. $C_{\text{disc}}(t)$

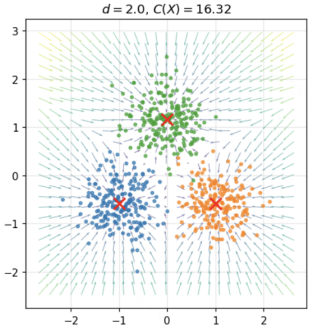
**Analytic kernel on $S^2$.**

$$h_t(x,y) = \sum_{\ell\geq 0} e^{-t\ell(\ell+1)}\,\frac{2\ell+1}{4\pi}P_\ell(\cos\angle(x,y)).$$

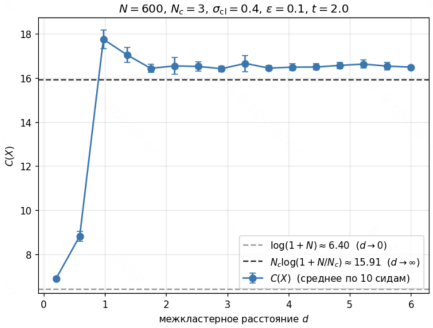
**Discrete operator.** Berry–Sauer double normalization of Gaussian affinities $\Rightarrow$ $H_t^{\text{disc}} = \exp(-t\,L_\epsilon^{\text{sym}})$, with $L_\epsilon^{\text{sym}} \to \Delta_{S^2}$ (Thm. 4.7). **Takeaway.** Curves $C_{\text{disc}}(t)$ and $C_{\text{analytic}}(t)$ overlap on the mesoscale, validating the discrete construction.

# Experiment 3 — Gaussian mixture in $\mathbb{R}^2$

**Goal.** Sensitivity of $C(X)$ to cluster separation $d$; analytical $\mathbf{s}(x) = \nabla \log p(x)$;




Three clusters with score field



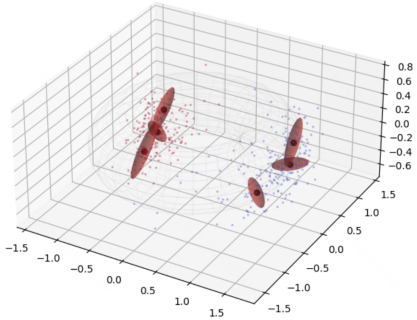

$C(d)$ vs. inter-cluster distance

**Result.**

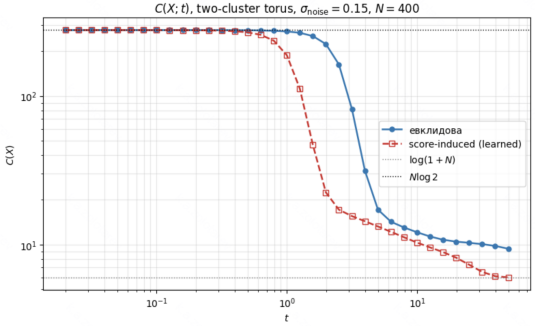
- **Growth in $d$.** $C(d)$ increases from a single-cluster value at $d \to 0$ to the plateau $N_c \log(1 + n/N_c)$ at $d \to \infty$; both limits match theory.
- **Indicator of separability.** The kernel sees one cluster at small $d$ and $N_c = 3$ independent blocks at large.

# Experiment 4 — Noisy torus in $\mathbb{R}^3$

**Setup.** Two clusters on a torus in $\mathbb{R}^3$ with ambient noise around the surface; score learned by a denoising network trained on the same distribution.
**Question.** Same kernel construction; does $C(X)$ reflect the geometry?




Torus with two clusters and learned score field




$C(X;t)$

**Result.** The curves diverge: the euc. kernel sees the 3-dimensional ambient volume including noise, while the score-induced kernel concentrates on the intrinsic 2-dimensional surface and resolves the cluster structure.

# Conclusion

## Contributions

- **Construction.** End-to-end pipeline: diffusion score $\Rightarrow$ Riemannian metric $g$ $\Rightarrow$ Berry–Sauer heat PSD kernel $\Rightarrow$ complexity $C(X) = \log\det(I+K)$.
- **Theory.** The score-induced kernel satisfies the moment conditions of Thm. 4.7 (Berry–Sauer): $L_\epsilon^{\mathrm{sym}} \to \Delta_g$. PSD on the sample yields invariance, monotonicity, subadditivity, submodularity, and Lipschitz stability.
- **Validation with experiments**

## Next steps

Realistic data (MNIST / CIFAR) with pretrained score networks; scalable computation of $K$;